Project Design Phase-II
Data Flow Diagram & User Stories

Date:	4 march 2026
Team ID:	SWTID-2026-2725
Project Name:	SpendWise
Maximum Marks:	4 Marks

Data Flow Diagrams:(Explanation)

Start → Home Page → Backend API (Node.js / Express) → Authentication → Expense Management → Database

i. User Interaction
   - The user accesses the SpendWise application through a web interface. The interface allows the user to register, login and manage expenses.

ii. Authentication Service
   - The system verifies the user's credentials. When a user logs in successfully, the system generates a secure authentication token to allow further access.

iii. Expense Controller
   - After authentication, the user can perform operations related to expense tracking such as adding new expenses, checking previous expenses, modifying existing records or deleting unnecessary records.

iv. Database Management

- All user information and expense details are stored in a database. The system retrieves and updates the data whenever the user performs any operation.

v. Response to User

- After each request, the system sends a response back to the user interface showing the updated information.

User Stories:

User Type	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
User (API User)	USN-1	As a user, I can register using name, email and password	Account is created and confirmation message shown	High	Sprint-1
User (API User)	USN-2	As a user, I can login using my email and password	System validates details and generates JWT token	High	Sprint-1
User (API User)	USN-3	As a user, I can securely access my dashboard	System allows access only with valid token	High	Sprint-1

User Type	User Story Number	User Story / Task	Acceptance Criteria	Priority	Release
User (API User)	USN-4	As a user, I can add a new expense record	Expense details are saved in database	High	Sprint-2
User (API User)	USN-5	As a user, I can view my expenses list	System displays all expenses belonging to the user	High	Sprint-2
User (API User)	USN-6	As a user, I can edit my expense details	Updated expense is saved successfully	Medium	Sprint-3
User (API User)	USN-7	As a user, I can delete an expense	Selected expense is removed from database	Medium	Sprint-3
User (API User)	USN-8	As a user, I cannot access other users' expenses	System restricts unauthorized data access	High	Sprint-3